Bloomberg P&L Data Extraction

Data Analytics Project Brief • Analysis and Visualization of P&L Data for **HEIA NA Equity**, FY2017–FY2024

Tools	Python, Excel, Power BI	Libraries	Pandas, NumPy, openpyxl
Dataset	8 fiscal years · 5 data sheets + README · EUR mn	Deliverable	Formatted P&L model + dashboard + memo

A Bloomberg terminal will give you an income statement in seconds, but what it gives you is not a report. In this project you are the analyst who pulled an eight-year income statement for the ticker **HEIA NA Equity** — a global brewer — and has been asked to turn it into a concise, decision-ready report. Brewing adds complications a retailer does not have: revenue is reported both *before* and *after* excise duty, volumes are measured in hectolitres, a pandemic closed the bars and restaurants that buy a large share of the product, raw-material and energy inflation hit the cost of every bottle, and the company exited a market part-way through the period. You will clean and restructure the extract in Python, rebuild it as a lookup-driven P&L model in Excel, format it to a professional standard, chart it, and finish with a Power BI dashboard. Finally you will interpret the firm's performance through trend, margin, volume and price analysis — and say clearly what the numbers do and do not support.

What to submit: (1) your Python notebook or script plus the cleaned dataset it produces, (2) your formatted Excel workbook containing the master P&L, the ratio analysis and the charts, (3) your Power BI (.pbix) file, and (4) a short write-up answering the questions below, ending with the one-page memo from Stage 9. Every number or claim you present must trace back to your own analysis — do not state figures you cannot reproduce.

Note on the data: the supplied extract is synthetic. It reproduces the structure, the reporting conventions and the untidiness of a real terminal download, and it is internally consistent once cleaned, but the figures are generated for teaching and must not be presented as actual company results.

STAGE 0 Project Goal & Problem Statement

1. In your own words, explain the goal of this project. Who is the decision-maker at the other end, and what decision does your output help them make?
2. Terminal data arrives raw. Explain why presentation quality is a genuine analytical risk and not merely cosmetic. Give one example of a structural or formatting flaw in a brewer's P&L that could lead a manager to the wrong conclusion.
3. State at least three objectives (concrete steps) that will lead you to the goal.
4. You will use Python, Excel and Power BI. Explain the role each tool plays in this workflow and why using all three is more useful than using any one of them alone.
5. The extract spans FY2017–FY2024. Name three events you would expect to be visible in the numbers over that window, and say which line of the P&L — or which revenue split — you would inspect first for each.
6. A brewer reports *Total Revenue* and *Net Revenue*. Before opening the file, explain what excise duty is and why an analyst would normally build margins on net revenue.

STAGE 1 Audit the Extract — Before You Change Anything

1. Open the workbook. List every sheet and describe in one line what each one holds. How many fiscal years and how many distinct line items are you dealing with?
2. On the raw extract sheet, where does the actual data table begin? What sits above it, and why can no tool read the file as-is?
3. The period labels occupy two rows. What goes wrong if you import this file assuming a single header row?
4. Which cells are not numbers even though they look like numbers? Find at least **five** distinct ways a value in this file has been made unusable for arithmetic. (Hint: not every space is a space, and not every dash is a minus sign.)
5. Negative numbers are written in more than one style on the same sheet. List every style you can find and the rows where each appears.

6 Find every placeholder that has been used in place of a value (there are four different ones across the workbook). Explain what each one means and how you intend to treat it — and why deleting them all would be a mistake.

7 Which rows are subtotals rather than line items, and which row appears twice? Why must you tag these before you aggregate anything?

8 One line item appears to have been extracted in a different unit from the one named in the sheet header. Identify it and prove it using the surrounding rows.

9 One deduction is shown with the opposite sign convention from every other cost on the statement. Identify it and show that *Total Revenue* only reconciles to *Net Revenue* once you account for it.

10 The total on the OPEX DETAIL sheet disagrees with the total on the raw income statement in two years. Find them and explain the difference.

11 Build a data-quality log with four columns: issue · sheet and row · evidence · your fix. You will refer back to this log in Stage 9.

STAGE 2 Data Cleaning & Restructuring (Python)

1 Read the extract (the CSV mirror or the workbook) with pandas. Why is starting with header=None (or an explicit skiprows) safer than letting pandas guess? Which encoding do you need, and why?

2 Write one function that converts the text values to numerics — handling comma and non-breaking-space thousands separators, padding spaces, parentheses negatives, leading minus signs (including the Unicode minus), percentage signs and placeholders. Show the function and four before/after examples, one of which must be a case that a naive float() call gets wrong.

3 Deal with the duplicated rows on the income statement and the segment sheet. How did you detect them, and how do you know a duplicate is a duplicate rather than a legitimate repeat?

4 Normalise the labels: strip whitespace, standardise case, and resolve names that differ between sheets (for example the digital channel, the Central & Eastern Europe region and the Asia-Pacific region). Which label pairs required a judgement call, and what did you decide?

5 Reshape the statement from wide (years across columns) to tidy long form — one row per line item per fiscal year. Why does tidy form make the downstream lookups and the Power BI model easier?

6 Correct the unit inconsistency and the sign-convention issue you found in Stage 1, and record both corrections in your log.

7 Write reconciliation checks in code. Assert that: revenue by channel sums to net revenue; total revenue minus excise equals net revenue; gross profit equals net revenue minus cost of sales; EBIT ties to its own build-up (gross profit + other income – operating expenses – impairment); income before taxes ties to EBIT plus the lines below it; net income attributable equals net income less minority interest; and EPS equals net income attributable to shareholders divided by weighted average shares (state your rounding tolerance). Report any year that fails and explain why.

8 Export a clean dataset for Excel and Power BI. What columns does it carry, and why those?

STAGE 3 Master P&L Table with Lookup Functions (Excel)

1 Build a master P&L sheet with fiscal years across the columns. Populate every cell using lookup functions against your cleaned data — XLOOKUP, or INDEX with MATCH — not by typing or pasting values. Show one formula and explain each argument.

2 Make it a two-way lookup, so any line item for any fiscal year can be retrieved by changing two inputs. Explain your approach.

3 Use the line-item dictionary on the LOOKUP TABLES sheet to pull each item's statement group and sign convention into your model. Two codes are missing from the dictionary — what does your formula return, and how did you make that failure graceful rather than ugly?

4 The dictionary labels do not match the extract labels perfectly ('&' versus 'and', British versus American spelling, trailing spaces). Which text functions did you need — and why does TRIM alone not remove every unwanted space in this file? Why does exact-match matter more than approximate-match here?

5 Regional segmentation was restated with effect from FY2021, and one old region was split across two new ones. Use the mapping table, including its allocation column, to build one comparable regional revenue view covering all eight years. State plainly what your restated series can and cannot support — and which two regions depend on the allocation assumption.

6 Why are lookup-driven cells safer than pasted values in a report that management will re-run next quarter?

STAGE 4 Trend Analysis

1 Compute year-over-year growth for net revenue, gross profit, operating profit and net income attributable to shareholders. Present it as one clean table, and decide how to show growth when the prior year is negative.

2 Compute the FY2017–FY2024 CAGR for net revenue and for net income attributable. Why do the two differ so sharply, and which is the more honest headline number?

3 Build an indexed series (FY2017 = 100) for net revenue, gross profit and total operating expenses on a single chart. What does the divergence between the three lines tell you?

4 Identify the best and the worst year for each of net revenue, operating profit and EPS. What appears to have happened in each case?

5 Over the full period, which grew faster — net revenue or operating expenses? What does your answer imply about operating leverage in this business?

6 One year's growth rate is distorted by a base effect. Identify it and explain how you would present that year fairly to a reader who only looks at the percentage.

7 Using the beer-volume row on the OPEX DETAIL sheet, compute net revenue per hectolitre for every year (the source leaves one year as n.a. — fill it yourself). Split the FY2019–FY2024 change in net revenue into a volume effect and a price/mix effect, in EUR millions.

STAGE 5 Profitability & Margin Analysis

1 Compute gross margin, operating margin and net margin (on net revenue) for all eight years and chart them together.

2 Recompute gross margin on *total* revenue including excise. How much lower is it, and which basis would you show management? Defend your choice.

3 Decompose the change in operating margin from FY2019 to FY2024 into gross margin, other income and each operating expense line. Which single line did most of the damage, and by how many percentage points?

4 Express every operating expense as a percentage of net revenue. Which line is the most volatile in absolute EUR, and which behaves most like a fixed cost? Why can a fixed cost look volatile when it is expressed as a percentage of sales? What does that mean for the firm in a downturn?

5 Between FY2021 and FY2023 net margin falls further than operating margin. Explain why, using the lines below EBIT — in particular the finance lines and discontinued operations.

6 Reconcile EPS back to net income attributable and the weighted average share count. How much of the EPS movement between FY2022 and FY2024 is profit, and how much is the share count?

7 In one year the effective tax rate exceeds 100%. Identify it, explain how that can arise, and how you would present that year's profitability without misleading the reader.

8 If you could put one ratio in front of management as the best measure of underlying performance here, which would it be? Defend your choice.

STAGE 6 Segment & Geographic Analysis

1 Show revenue mix by product category and by distribution channel for FY2017 and FY2024 side by side, and by brand for FY2017 and the last year brand revenue is available. What shifted, and by how much?

2 Compare on-trade and off-trade: compute each channel's share of net revenue and its growth over the period. What does the shift suggest about how the business goes to market — and how much of it reversed after FY2020?

3. The digital B2B channel shows by far the highest growth rate. Why is that percentage misleading on its own, and what figure would you report alongside it?
4. Brand-level revenue stops after FY2022. Find the reason in the extract's own footnotes and explain how you handled the gap — and how you disclosed it.
5. On your restated regional basis, which region contributed most of the growth and which most of the decline? Compute each region's contribution to the total change in net revenue from FY2019 to FY2024, in EUR millions. Why is a contribution analysis more useful to management than a table of growth rates?
6. Compute net revenue per employee and net revenue per hectolitre for each year and comment on the two trends together. What are the limits of each as an indicator of productivity?

STAGE 7 Professional Formatting & Excel Charts

1. Format the master P&L to a standard a managing director would accept without comment: consistent number formats, a clear convention for negatives, units named in the header, aligned decimals, no text-formatted numbers left anywhere, sensible column widths, frozen panes, and a print area set to one page wide.
2. Which conventions did you adopt to distinguish inputs, formulas and cross-sheet links? Why does that matter in a model somebody else will open six months from now?
3. Apply conditional formatting so margin direction is visible at a glance. State what a reader should be able to see in under five seconds.
4. Build at least five charts and justify the chart type for each: net revenue with an operating-margin overlay (combo), the three margins over time (line), operating expenses as a share of net revenue (stacked), regional revenue on the restated basis (bar), and the volume versus price/mix bridge from Stage 4 (waterfall).
5. What did you deliberately leave off your charts, and why? Consider gridlines, decimals, dual axes, legends and labels.
6. Add a one-page summary sheet — the numbers and charts management should see first, and nothing else.

STAGE 8 Power BI Dashboard

8.1 Model and dynamic filters (slicers)

Load your cleaned dataset and add interactive filters so a user can slice by:

- Fiscal year (and a year range)
- Statement group — revenue, revenue deduction, cost, operating expense, financial, tax, subtotal
- Reporting region (on your restated basis)
- Product category and brand
- Distribution channel

8.2 Key visuals and KPIs

Build visuals that answer each of the following, choosing an appropriate chart type for each and justifying it:

- Net revenue and revenue growth KPI cards, with the latest year against the prior year
- The margin trio — gross, operating and net — over the full eight years
- A waterfall from total revenue (incl. excise) down to net income attributable for the latest fiscal year
- Indexed trend of net revenue against total operating expenses (FY2017 = 100)
- Net revenue by region, on the restated comparable basis
- Operating expense composition as a share of net revenue
- Beer volume against net revenue per hectolitre by year
- EPS and dividend per share by year
- Insight cards stating a key takeaway in plain language

1. Explain your data model. Did you load one tidy fact table or several tables with relationships (for example a line-item dimension built from the dictionary and a region dimension built from the mapping table)? Justify it.
2. Which measures did you write in DAX rather than pre-computing in Python, and why does that division make sense?

3 Explain your layout choices. How does the page read top to bottom for a reader who has sixty seconds?

4 Which single visual is most useful to a CFO, and which to an external equity analyst? Are they the same one? Explain.

STAGE 9 Insights & Storytelling

Use your model and dashboard to answer the following. Report the actual figures you find.

1 In which year did net revenue peak, at what level, and what was the total change from FY2017 to FY2024 in EUR millions and as a CAGR?

2 Which year had the highest gross margin and which the lowest? What is the spread in percentage points?

3 Which year saw the largest single-year fall in operating profit, and by how much in EUR millions? Which year saw the second largest, and what was different about its cause?

4 Which operating expense line rose most as a share of net revenue across the period, and by how many points?

5 Which distribution channel and which region grew fastest over the eight years? For the channel, state both the growth rate and the absolute increase.

6 Which year delivered the highest EPS and which the lowest? What is the range? Was a dividend still paid in the lowest year, and what does that tell you?

7 Was the weakness in FY2022–FY2023 a revenue problem, a margin problem, or a below-the-line problem? Defend your answer with numbers.

8 Write three short insight cards — one or two sentences each — that a non-finance manager could understand immediately.

9 Write a one-page memo to the CFO: what the eight-year record says about this business, the two numbers you would watch next year, and one recommendation you can support from your own analysis.

Final reflection: Look back at the data-quality log you started in Stage 1. Which single defect in the raw extract, if you had missed it, would have most changed your conclusions — the unit error, the excise sign convention, the impairment hidden in one opex total, or something else? Then explain how combining Python (cleaning and reconciliation), Excel (structure, lookups and formatting) and Power BI (storytelling) let you reach a defensible, data-driven view rather than an impression.